

MARK LORENZ YAMANAKA

Aeronautical Engineer | Aerospace Structural Design and Analysis (CATIA, FEA, Hand Calculations)

Tsukuba, Ibaraki, Japan | +81-70-3276-7554 | Japanese national (no visa required) | Open to relocation
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SUMMARY

PRC-licensed Professional Aeronautical Engineer (2024) focused on aerospace structural design and analysis. Project work spans CFD, FEA, structural sizing and GD&T drawing packages, plus maintenance and ground-support internships across A320, ATR 72, B737 and Cessna 172. Analysis practice is validation-first: hand calculations set the baseline, mesh convergence is shown, and results are published in full, including those that fell short of target. Japanese national, raised and educated in the Philippines.

PROJECTS

Uncooled SS316 Convergent-Divergent Nozzle | [Link](#)

- Carried an uncooled SS316 C-D nozzle ($P_c = 2$ MPa, $T_c = 800$ K) from CFD through one-way coupled thermal-structural FEA into a GD&T drawing set; linearised membrane-plus-bending stress placed the governing location at the chamber-to-converging junction (29.6 MPa), not the throat, giving yield FoS 7.12 across a three-level mesh convergence study.
- Passed CFD checks at peak wall $y^+ 1.74$, discharge coefficient 0.977 and exit Mach 2.85 against 2.94 by hand calculation; identified creep at a ~ 527 °C wall, not static yield, as the real design driver.
- Sized an 8× M8 A286 bolted joint per ASME BPVC VIII-1 Appendix 2 and detailed the package in CATIA V6 (CDN-000 to CDN-002) with tolerance stacks closing at 1.68×, machining route, inspection plan and BOM; a self-review against ASME Y14.5-2018 logged 5 blocking findings, revised after external peer review.
- Published the v1.0 error (FoS 42.70, traced to a wrong assessment location and stress misclassification) alongside the corrected v2.0.

AI 6061-T6 Propellant Tank: Static FEA Verification | [Link](#)

- Ran static linear FEA of a 3 mm-wall tank with hemispherical heads at 3.25 MPa (inner radius 147 mm, $R/t = 49$) in CalculiX via PrePoMax (C3D10); mid-cylinder von Mises 136.6 MPa against 137.91 MPa by thin-wall hand calculation (1.0%), with a 0.62% mesh change at 264,007 nodes.
- Reported FoS 1.91 against a self-defined 2.0 target at the cylinder-to-head junction, a 3D discontinuity effect thin-wall theory does not capture, and recommended a 3.2 mm wall; documented a boundary-condition correction that moved constraints off the junction (peak 204.1 to 144.5 MPa).

MK-10 Light Aircraft: FAR Part 23 Design and Structural Sizing | [Link](#)

- Held one design baseline from mission definition through weight and CG, drag buildup and the V-n envelope for a two-seat light aircraft (1,649 lb gross weight, 126.5 ft² wing, aspect ratio 6.58, NACA 2415, 125 BHP); verified 459 nmi range, 7.63 h endurance and 9,536 ft service ceiling.
- Sized front and rear wing spars by hand calculation per Bruhn from a 10-strip spanwise load distribution through shear and moment diagrams, then resolved control surfaces, landing-gear load cases and a SolidWorks three-view.

PANO-VISTA: Fixed-Wing UAV Design and Systems Integration | [Link](#)

- Five-person team project: modelled a tractor-configuration fixed-wing UAV in one SolidWorks master assembly and built it with brushless propulsion (A2212/6T 2200KV, 40A ESC, LiPo), an eight-channel receiver and four servos across three control axes; delivered exploded and orthographic drawings with BOM.

Two-Blade Wooden Rotor: Drawing Package | [Link](#)

- Drafted a $\varnothing 71.10$ in two-blade rotor by hand in orthographic projection, then rebuilt it parametrically in SolidWorks as a blade-and-hub assembly with dimensioned three-view; twist falls from 78° inboard to 24° at the tip; solidity 0.095.

High-Lift Device Demonstrator: Servo-Actuated Wing Section

- Built a sectioned wing (cardboard ribs, aluminium-foil skin) with servos between the ribs driving ailerons and leading-edge slats from an Arduino Uno sketch in C++, operated from a base-mounted control interface.

INTERNSHIPS

Aircraft Maintenance Engineer Intern | Dornier Technology Inc.

May 2023 – Jul 2023

- Supported 10+ base and line maintenance checks at a third-party MRO on customer fleets (Cebu Pacific, Royal Air): A320, ATR 72-500, B737-200 and BAe 146, under a PCAR Part 6 AMO (FAR Part 145 equivalent).
- Performed pre- and post-check inspections in line maintenance; assisted access panel removal and refit and fastener removal and installation.
- Inspected in-service airframe structure for damage and corrosion; observed and assisted dye penetrant and borescope inspection.
- Handled the records chain for each task (task cards, work orders, non-routine cards, certificate of release to service); documentation closed an internal audit with zero non-conformities. Reported to licensed aircraft maintenance engineers and base maintenance supervisors.

Ground Support Group Intern | Alpha Aviation Group Philippines

Feb 2023

- Trained under flight instructors on aircraft systems and flight school operations, including Cessna 172 and A320 simulator sessions.
- Performed external inspections, equipment checks and cockpit preparation on trainer aircraft, and a supervised Cessna 172 engine run-up and taxi with function and fault checks.

WORK EXPERIENCE

Part-Time Leader / Sub-Leader (Rotating Shift Lead) | ZOZOBASE

Feb 2018 – May 2026

- Rotated as shift lead for a 15-member team in a Japanese-language workplace, running the packaging, transportation and logistics workflows for each shift.
- Identified a bottleneck in packaging transport and proposed a revised workflow to the manager; escalated to supervisors and adopted, raising throughput 20% and cutting cycle time 12%.
- Requested daily order reports from other departments to scope each shift and set team priorities; reported shift output to the manager at close.
- Briefed and trained new recruits on the workflows and integrated them into the team.

LEADERSHIP

Chairperson / Officer-in-Charge | SEA-HAU Debate Committee

Sep 2022 – Nov 2022

- Led a university-wide debate program with 25+ participants, coordinating judges and partner organizations from planning through execution.
- Developed standardized evaluation criteria and topic frameworks for fair, consistent judging.

Point-Person | SEA-HAU Debate Committee

Jul 2022 – Aug 2022

- Acted as primary liaison between participants and the organizing committee; guided participants on parliamentary debate procedure and coordinated teams, adjudicators and committee members during events.

EDUCATION

Bachelor of Science in Aeronautical Engineering

Jul 2020 – May 2024

Holy Angel University, Angeles City, Philippines

- Honors: Dean's List (GPA 3.54/4.0)
- Relevant Coursework: Aircraft Structures, Aircraft Systems, Aircraft / Helicopter / UAV Design, Aerodynamics, Powerplant, Reliability Engineering, Engineering Data Analysis

SKILLS AND CERTIFICATIONS

- **Stress and Structural Analysis:** Hand calculations, stress linearisation, membrane-plus-bending classification, V-n envelope, limit and ultimate load, mesh convergence, sensitivity studies
- **FEA and CFD:** ANSYS Mechanical, ANSYS Fluent, one-way FSI, CalculiX, PrePoMax
- **CAD and Drawings:** CATIA V6, SolidWorks, ANSYS SpaceClaim, GD&T, tolerance stack-up, BOM, machining route, inspection plans
- **Standards:** FAR Part 23, ASME Y14.5-2018, ASME BPVC VIII-1 Appendix 2, PCAR Part 6, FAR Part 145
- **Documentation:** Analysis reports, task cards, work orders, non-routine cards, certificate of release to service
- **Programming and Tools:** Python, C++, Git/GitHub, MATLAB, Microsoft Office
- **Certification:** PRC Professional Aeronautical Engineer License, Philippines (Dec 2024; board exam, 86% average)

LANGUAGES

- **English:** Native | **Japanese:** Beginner (reads kana, no JLPT yet) | **Cebuano:** Native | **Filipino:** Professional